

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Justification

Input content is the highest-priority dimension per the elicitation (HIGH) and shows the most significant validity concerns. While MILU is India-first and includes some North Indian state-specific content (UP minerals, Rajasthan geography, Chikankari, Panchayati Raj), the dataset analysis reveals a CRITICAL finding: 100% of the 245 sampled Hindi validation items carry is_translated: True, far exceeding the paper's reported dataset-wide 25% translation rate. This raises serious concerns about Hindi-register fidelity for items used as few-shot exemplars. Multiple MAJOR concerns compound this: canonical Hindi literary figures (Tulsidas, Premchand, Kabir, Mahadevi Varma) were entirely absent from the sample; English vocabulary questions ('didactic', 'grim', 'Evangelize') appear inside Hindi Language Studies; ~3-4% of items have truncated stems making them unevaluable; framing is dominated by UPSC/SSC/state-civil-service GK rather than board syllabus content. State-wise exam-source distribution for Hindi items is undocumented. The deployment user did note that competitive-exam answer keys are broadly compatible with North Indian teacher norms, mitigating but not eliminating these concerns.

Strengths

- Genuine North Indian state-specific content present (UP minerals, Rajasthan districts and literacy, Bihar e-cigarette policy, UP Chikankari) [DATASET-D6, DATASET-D15, DATASET-D26, DATASET-D34].
- India-first design with explicit inclusion of regional and state-level exam material [Q11, Q14, Q28].
- Hindi grammar items (indirect speech, postpositions, voice) align with board Language Studies content [DATASET-D4, DATASET-D20].
- Indian constitutional and Panchayati Raj content aligned with civics syllabi [DATASET-D2, DATASET-D19, DATASET-D48].

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — deployment requires Hindi-belt cultural, regional, and dialectal knowledge (canonical Hindi literature, North Indian medieval history, state administrative content). MILU includes some such material [Q2, Q12, DATASET-D6, DATASET-D34] but coverage is uneven.

IC-2: Partial alignment. India-centric framing is genuine [Q11, Q30] but the validation sample showed canonical Hindi-board literary figures absent and English-vocabulary items embedded in Hindi Language Studies subject [DATASET-D16, DATASET-D18, DATASET-D22], indicating cultural/curricular misalignment for the target deployment.

ID	Evidence
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WEB-19	Bihar's Hindi-as-mother-tongue is only ~25.5% — relevant to whether Bihar Hindi-medium teachers' register expectations match annotator outputs.
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IC-3: Some Western/non-regional content present: William Wordsworth, Man Booker Prize, English vocabulary items [DATASET-D3, DATASET-D25, DATASET-D16]. Also South Indian regional content (Pandya, Silappatikaram, Tamil Bharati) [DATASET-D10, DATASET-D23, DATASET-D33] — appropriate for pan-India exams but dilutive for North-Indian-board deployment.

IC-4: INSUFFICIENT DOCUMENTATION — MILU does not document recruitment of regional annotators by language; AI4Bharat volunteers and online-portal subject experts are referenced [Q26, Q87] but no Hindi-medium North Indian regional annotator pool is specified. Web search confirmed no public source resolves this (gap_id 6).

IC-5: Multiple content issues compromising content validity: (a) 100% translation rate in the sampled validation split [DATASET-D35], (b) absence of canonical Hindi literary canon from sample, (c) truncated/unanswerable items (~3-4%) [DATASET-D37 through D44], (d) competitive-exam framing dominance [Q24, DATASET-D17, DATASET-D47], (e) English vocabulary masquerading as Hindi Language Studies [DATASET-D16, DATASET-D22].

ID	Evidence
D35	All 245 examples have `is_translated: True` — Every sampled item is marked as translated from English

Information Gaps

- Hindi-specific translation proportion in test split unknown; whether the 100% translation rate seen in validation extends to test split needs gated-access verification.
- State-wise (UP/MP/Rajasthan/Bihar) exam-source distribution for Hindi items not published.
- No comparison published between MILU Hindi items and UP/MP/Rajasthan/Bihar board syllabi.
- Whether GPT-4O Hindi translations conform to Vaigyaanik evam Takniki Shabdavali Aayog standardised scientific terminology vs. anglicised transliteration.

Requires Expert Verification

- Hindi-medium board teacher review of a sample of translated items for register fidelity.
- Subject-expert audit of truncated items and items where English vocabulary is tested inside Hindi Language Studies.
- Test-split sampling to confirm whether canonical Hindi literary figures appear with adequate coverage.

Remediation

Gap 1: 100% of sampled Hindi validation items are translated (vs. paper's claimed 25% dataset-wide); canonical Hindi-board literary figures absent from the sample; English vocabulary embedded in Hindi Language Studies.

Recommendation: Filter MILU Hindi items by is_translated flag and prefer natively sourced items for evaluation; supplement with a custom Hindi Literature MCQ set covering canonical board authors (Tulsidas, Premchand, Kabir, Mahadevi Varma); reclassify or remove English-vocabulary items mislabelled under Hindi Language Studies.